Algebraic

Week 1B

Overview

Expansion

Key Ideas

- *Expanding using Distributive Law Twice*
- *Expanding Binomial Products*
- *Further Expansions*
- *Pascal Triangle*

Key Idea - Expanding using the Distributive Law Twice

The expression $(a + b)(c + d)$ is called a binomial product because two binomial expressions are multiplied together. The expansion of this product requires use of the **Distributive Law twice.**

$$\mathbf{(a + b)(c + d) = a\,(c + d) + b(c + d)}$$

$$\mathbf{= ac + ad + bc + bd}$$

Example,

Expand and simplify

$$(x + 2)(2x - 1) = x(2x - 1) + 2(2x - 1)$$

$$= x \times 2x + x \times (-1) + 2 \times 2x + 2 \times (-1)$$

$$= 2x^2 - x + 4x - 2$$

$$= 2x^2 + 3x - 2$$

Exercise: Expand and simplify each of the following expressions:

(a) $(2m+2)(3m-4)$ $= 2m(3m-4)+2(3m-4)$

$= 6m^2-8m+6m-8$

$= 6m^2-2m-8$

(b) $(a-7)(3a^2-a)$ $= a(3a^2-a)-7(3a^2-a)$

$= 3a^3-a^2-21a^2+7a$

$= 3a^3-22a^2+7a$

Key Idea - Expanding Binomials Products

There are two types of quadratic polynomial that we will consider here:

Square of a binomial

and

Expansions resulting in Difference of two Squares.

Square of a binomial

Square of a binomial can either be expanded by using Pascal's triangle or using the distributive law.

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b)\\ &\ a(a+b)+b(a+b)\\ &= a^2+ab+ba+b^2\\ &= a^2+2ab+b^2 \quad \text{since } ab=ba\end{aligned}$$

Likewise,

$$\begin{aligned}(a-b)^2 &= (a-b)(a-b)\\ &= a(a-b)-b(a-b)\\ &= a^2-ab-ba+b^2\\ &= a^2-2ab+b^2 \quad \text{since } ab=ba\end{aligned}$$

Example:

Expand the following:

$$(a+3)^2 = a^2 + 3a + 3^2$$
$$= a^2 + 3a + 9$$

$$(3x-2y)^2 = (3x)^2 - 2(3x)(2y) + (2y)^2$$
$$= 9x^2 - 12xy + 4y^2$$

$$(mn+p)^2 = (mn)^2 + 2mnp + (p)^2$$
$$= m^2n^2 + 2mnp + p^2$$

Exercise: Expand and simplify:

(a) $(3a-4)^2$

$= (3a)^2 - 2\times 3a \times 4 + 4^2$

$= 9a^2 - 24a + 16$

or $(3a-4)(3a-4)$

$= 3a(3a-4) - 4(3a-4)$

$= 9a^2 - 12a - 12a + 16$

$= 9a^2 - 24a + 16$

(b) $(3xy+z)^2$

$= (3xy)^2 + 2\times 3xy \times z + z^2$

$= 9x^2y^2 + 6xyz + z^2$

(c) $-3(2x^3+2)^2 + x^3$

$= -3\left((2x^3)^2 + 2\times 2x^3 \times 2 + 2^2\right) + x^3$

$= -3\left(4x^6 + 8x^3 + 4\right) + x^3$

$= -12x^6 - 24x^3 - 12 + x^3$

$= -12x^6 - 23x^3 - 12$

Expansion resulting in Difference of Two Squares

Another important binomial product that is of the form $(a - b)(\mathrm{a} + \mathrm{b})$, has a unique expansion called the ***difference of two squares because:***

$$\begin{aligned}(a-b)(a+b) &= a(a+b)-b(a+b)\\ &= a^2+ab-ba+b^2\\ &= a^2-b^2\end{aligned}$$

Rather than expanding expressions of this type by applying the Distributive Law, ***this result should be memorised:***

$$(a-b)(a+b)=a^2-b^2$$

Expressions like

$$x^2-4,\quad a^2-b^2,\quad 9y^2-16,\quad 1-4a^2$$

are all examples of expression involving the difference of two squares.

For the examples below we can apply the Distributive Law. However, it is quicker to notice that the terms in each set of the brackets are the same, the only difference being the sign in the middle. We notice that the answer will be the difference of two squares.

***Examples*:**

Expand the following:

$$\begin{aligned}(x-2)(x+2) &= x^2 - 2^2 \\ &= x^2 - 4\end{aligned}$$

$$\begin{aligned}(3y-4)(3y+4) &= (3y)^2 - 4^2 \\ &= 9y^2 - 16\end{aligned}$$

$$\begin{aligned}(1-2a)(1+2a) &= 1-(2a)^2 \\ &= 1-4a^2\end{aligned}$$

$$\begin{aligned}(2x-y^2)(2x+y^2) &= (2x)^2 - (y^2)^2 \\ &= 4x^2 - y^4\end{aligned}$$

$$\begin{aligned}(2-\sqrt{x})(2+\sqrt{x}) &= (2)^2 - (\sqrt{x})^2 \\ &= 4 - x\end{aligned}$$

Exercise: Expand and simplify:

(a) $(5-2y)(5+2y)$ $= 5(5+2y)-2y(5+2y)$

$= 25+\cancel{10}y-\cancel{10}y-4y^2$

$= 25-4y^2$

easier to learn that it expand to a difference of 2 squares

$5^2-(2y)^2=25-4y^2$

(b) $(7x-2y-3x)(4x+4y-2y)$

$(4x-2y)(4x-2y) = (4x)^2-(2y)^2$

$= 16x^2-4y^2$

(c) $(2x-3)(2x-3)+(x-1)(x+1)$ $= 4x^2-6x-6x+9+x^2-1$

$= 5x^2-12x+8$

Exercise: Are any of these expressions equal? Explain your answer!!!!!!

$$(a+b)^2, \quad (a-b)^2, \quad a^2-b^2$$

$(a-b)(a+b)$

$a^2+2ab+b^2$

$a^2-2ab+b^2$

No expressions are equal as none of them has the same expansion

Key Idea - Further Expansions

For products involving expansions of expressions with more than two terms, we apply the **Distributive Law repeatedly.** This method also applies when multiplying more than two binomial factors.

***Examples*:**

Expand and simplify the following:

$$\begin{aligned}(3x-4)(x^2-2x+5) &= 3x(x^2-2x+5)-4(x^2-2x+5)\\ &= 3x^3-6x^2+15x-4x^2+8x-20\\ &= 3x^3-10x^2+23x-20\end{aligned}$$

$$\begin{aligned}(3-x)(2-x)(x+1) &= \left[3(2-x)-x(2-x)\right](x+1)\\ &= (6-3x-2x+x^2)(x+1)\\ &= (6-5x+x^2)(x+1)\\ &= 6(x+1)-5x(x+1)+x^2(x+1)\\ &= 6x+6-5x^2-5x+x^3+x^2\\ &= x^3-4x^2+x+6\end{aligned}$$

Exercise: Expand the following:

(a) $(2b-1)(a+2ab+b)+2(a-3ab)$ $= 2b(a+2ab+b)-1(a+2ab+b)+2(a-3b)$

$= \cancel{2}ab+4ab^2+2b^2-a-\cancel{2}ab-b+2a-6ab$

$= -6ab+a+4ab^2+2b^2-b$

(b) $5(3b-1)^2(b+2)$ $= 5\left((3b)^2-2\times 3b\times 1+1^2\right)(b+2)$

$= 5(9b^2-6b+1)(b+2)$

$= 5\left[b(9b^2-6b+1)+2(9b^2-6b+1)\right]$

Or $5\left[9b^2(b+2)-6b(b+2)+1(b+2)\right]$

$= 5\left[9b^3-6b^2+b+18b^2-12b+2\right]$

$= 5\left[9b^3+12b^2-11b+2\right]$

$= 45b^3+60b^2-55b+10$

Key Idea - Using Pascal's Triangle for Expanding $(a+b)^n$

A particularly important type of product we often need to expand is

$$\boldsymbol{(a+b)^n}$$

The first few expansions are very simple:

$$(a+b)^0 = 1$$

because anything raised to the power 0 is 1.
Also

$$(a+b)^1 = a+b$$

because anything raised to the power 1 is itself.
Next

$$\begin{aligned}(a+b)^2 &= (a+b)(a+b)\\ &= a^2+ab+ba+b^2\\ &= a^2+2ab+b^2\end{aligned}$$

We'll do one more, just to see that it is getting complicated:

$$\begin{aligned}(a+b)^3 &= (a+b)(a+b)(a+b)\\ &= (a+b)\left(a^2+2ab+b^2\right)\left(\text{ we worked out} (a+b)^2 = a^2+2ab+b^2 \text{ above}\right)\\ &= a^3+2a^2b+ab^2+ba^2+2ab^2+b^3\\ &= a^3+3a^2b+3ab^2+b^3\end{aligned}$$

Okay, that's hard work. We don't want to have to do it every time. So, we use ***Pascal's Triangle***, whose nth row describes the coefficients that appear when we expand $(a+b)^n$.

Here are rows $0,\ 1,\ 2,\ 3,\ 4,\ 5,\ 6$ of Pascal's triangle (it goes on down forever):

$$\begin{array}{ccccccccccccc}
&&&&&&1&&&&&&\\
&&&&&1&&1&&&&&\\
&&&&1&&2&&1&&&&\\
&&&1&&3&&3&&1&&&\\
&&1&&4&&6&&4&&1&&\\
&1&&5&&10&&10&&5&&1&\\
1&&6&&15&&20&&15&&6&&1\\
&&&&&&\vdots&&&&&&
\end{array}$$

Notice that the top row or the triangle is the zeroeth row, not the first row. It's easy to find the nth row: it's the one that starts $1, n\ldots$. For example, the third row is

$$1 \quad 3 \quad 3 \quad 1.$$

The ***pattern*** when describing Pascal's triangle is that the entries down the sides of the triangle are all 1's, and each of the internal entries is equal to the sum of the two neighbouring entries in the row above. For example, looking in the triangle below, the neighbouring entries in the row above the circled position are the 4 and 6 in squares above it. These add up to 10, so the entry that goes in the circle is 10.

$$
\begin{array}{ccccccccccccc}
 & & & & & & 1 & & & & & & \\
 & & & & & 1 & & 1 & & & & & \\
 & & & & 1 & & 2 & & 1 & & & & \\
 & & & 1 & & 3 & & 3 & & 1 & & & \\
 & & 1 & & \boxed{4} & & \boxed{6} & & 4 & & 1 & & \\
 & 1 & & 5 & & \bigcirc & & 10 & & 5 & & 1 & \\
1 & & 6 & & 15 & & 20 & & 15 & & 6 & & 1
\end{array}
$$

When we expand out $(a+b)^n$, the nth row (that is, the one starting $1\ n\ldots$ OR the $(n+1)$th line) tells us the coefficients of the terms.

The terms start with a^n. Its coefficient is a 1 because 1 is the first entry in the nth row of the triangle. Then as we move across, the powers of a decrease and the powers of b increase. In each entry, the power of a and the power of b add up to n.

Example:

Use Pascal Triangle to expand $(a+b)^4$

1. We notice that we will have (4 + 1) = 4 terms as n = 5.
2. We then pick out the third row of the Pascal triangle (fourth line) and the coefficients are: 1 4 6 4 1. These will be the coefficients of a^4b^0, a^3b^1, a^2b^2, a^1b^3, a^0b^4.
3. The powers of the a term starts at 4 and decreases to 0.
4. The powers of the b term starts at 0 and increases to 4.
5. So, we obtain the following expansion. (Recall that $(a)^0=(b)^0=1$).

$$\begin{aligned}(a+b)^4 &= 1\times a^4\times b^0+4\times a^3\times b^1+6\times a^2\times b^2+4\times a^1\times b^3+1\times a^0\times b^4\\ &= a^4+4a^3b+6a^2b^2+4ab^3+b^4\end{aligned}$$

We will always have terms like a^nb^0 and a^0b^n, since any number to the power 0 is 1, we always write these just as a^n and b^n. Likewise, we will always just write a and b instead of a^1 and b^1 where they occur.

Example,

Use Pascal Triangle to expand $(1+x)^6$.

1. There are (6 + 1) = 7 terms as n = 6.

2. The sixth row in Pascal triangle (seventh line) provides the coefficients for this expansion: 1 6 15 20 15 6 1.

3. The 'a' term is 1 with the powers of 1 starting at 6 and decreasing to 0.

4. The 'b' term is x with the powers of b starting at 0 and increasing to 6.

5. Write the expression, evaluating the coefficients, and simplify where possible. Recall that $1^0 = 1$ & $x^0 = 1$

$$\begin{aligned}(1+x)^6 &= 1\times(1)^6\times(x)^0 + 6\times(1)^5\times(x)^1 + 15\times(1)^4\times(x)^2 + 20\times(1)^3\times(x)^3 + 15\times(1)^2\times(x)^4 + 6\times(1)^1\times(x)^5 + 1\times(1)^0\times(x)^6 \\ &= 1\times1\times1 + 6\times1\times x + 15\times1\times x^2 + 20\times1\times x^3 + 15\times1\times x^4 + 6\times1\times x^5 + 1\times1\times x^6 \\ &= 1 + 6x + 15x^2 + 20x^3 + 15x^4 + 6x^5 + 1x^6\end{aligned}$$

Exercise:

(a) Calculate 6th and 7th row of Pascal triangle.

1 —— 1st row $(a+b)^0$

1 1 —— 2nd row $(a+b)^1$

1 2 1 —— 3rd

1 3 3 1 —— 4th

1 4 6 4 1

1 5 10 10 5 1 ← 6th row $(a+b)^5$

1 6 15 20 15 6 1 ← 7th row $(a+b)^6$

(b) Use Pascal's triangle to write down the expansion of $(2+x)^5$

1. $5+1=6$ terms 2. coefficients are 1 5 10 10 5 1

$1\times 2^5\times x^0 + 5\times 2^4\times x^1 + 10\times 2^3\times x^2 + 10\times 2^2\times x^3 + 5\times 2^1\times x^4 + 1\times 2^0\times x^5$

$= 32 + 80x + 80x^2 + 40x^3 + 10x^4 + x^5$

Example,

Use Pascal Triangle to write down the expansion for $(1-2x)^3$.

1. There are (3 + 1) = 4 terms as $n = 3$.
2. The coefficients in the binomial expansion of this power have values from the 4th line of Pascal's Triangle, that is: 1 3 3 1.
3. The 'a' terms is 1 with the powers of 1 starting at 3 and decreasing to 0.
4. The 'b' terms is $(2x)$ with the powers of $(2x)$ starting at 0 and increasing to 3.
5. Write the expression, evaluating the coefficients, and simplify where possible.

Recall that $(1)^0 = 1$ & $(2x)^0 = 1$

$$
\begin{aligned}
(1-2x)^3 &= 1\times(1)^3\times(2x)^0 - 3\times(1)^2\times(2x)^1 + 3\times(1)^1\times(2x)^2 - 1\times(1)^0\times(2x)^3 \\
&= 1\times1\times1 - 3\times1\times2x + 3\times1\times4x^2 - 1\times1\times8x^3 \\
&= 1 - 6x + 12x^2 - 8x^3
\end{aligned}
$$

Note: *In his case the sign between the terms alternates, starting with "minus".*

Example,

Use Pascal Triangle to write down the expansion for $(1-2x)^3$. *An alternative way of expanding*

Firstly rewrite $(1-2x)^3$ as $(1+(-2x))^3$

1. There are (3 + 1) = 4 terms as $n = 3$.
2. The coefficients in the binomial expansion of this power have values from the 4th line of Pascal's Triangle, that is: 1 3 3 1.
3. The 'a' terms is 1 with the powers of 1 starting at 3 and decreasing to 0.
4. The 'b' terms is $(-2x)$ with the powers of $(-2x)$ starting at 0 and increasing to 3.
5. Write the expression, evaluating the coefficients, and simplify where possible.

Note: *That an odd power of* (-2x) *gives a negative term and an even power of* (-2x) *gives a positive term.*

So,

$$\begin{aligned}(1-2x)^3 &= 1\times(1)^3\times(-2x)^0+3\times(1)^2\times(-2x)^1+3\times(1)^1\times(-2x)^2+1\times(1)^0\times(-2x)^3\\ &= 1\times1\times1+3\times1\times(-2x)+3\times1\times4x^2+1\times1\times(-8x^3)\\ &= 1-6x+12x^2-8x^3\end{aligned}$$

Exercise: Use Pascal's triangle to write down the expansion of $(2y-x)^4$

✓ keep 2y in brackets i.e (2y)

↑ since it is a minus the sign alternates starting with minus

1. There are 4+1=5 terms
2. The coefficients are: 1 4 6 4 1

note: the x-term increases in power from 0 to 4

note: the (2y) term decreases in power from 4 to 0

$(2y-x)^4$

$$= \underline{1}\times\boxed{(2y)^4}\times\boxed{x^0}-\underline{4}\times\boxed{(2y)^3}\times\boxed{x^1}+\underline{6}\times\boxed{(2y)^2}\times\boxed{x^2}-\underline{4}\times\boxed{(2y)^1}\times\boxed{x^3}+\underline{1}\times\boxed{(2y)^0}\times\boxed{x^4}$$

$$= 16y^4-32y^3x+24y^2x^2-8yx^3+x^4$$